Wilson Harper

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Education

Rice University Class of 2029

- B.S. Electrical Engineering; Minor in Entrepreneurship
- Lillian Haynie Trustee Scholar; Martel College Outstanding New Student Award

Experience

Electrical Engineering Intern, RTX BBN Technologies Summer 2026

- Evaluated system architecture and performance for a low-cost mobile embedded system to optimize latency
- Designed a 4-layer PCB in Altium for harsh environments, integrating high-speed ICs and a robust PDN
- Field-tested radar detection against autonomous and manual mobile platforms across multi-site field trials
- Executed trade studies to select off-the-shelf boards and positioning systems under tight SWaP-C constraints
- Developed rugged field deployment solutions for high-speed, FPGA-based fiber-optic hardware systems

Mission Control Lead, Rice Eclipse Rocketry 2025 - Present

- Leading development of electronics and software for Rice's liquid engine development program
- Executed the program's first successful liquid hotfire, enabling the transition to future liquid launches
- Integrated LabJack DAQ and Raspberry Pi to sample sensors at 300 Hz, actuate solenoids, and ignite engine
- Designed avionics architecture for an active flight stabilization system intended for a spaceshot rocket
- Implemented RP2350 with I2C/UART/SPI sensors on a custom PCB to actuate servo canards in real-time

Undergraduate Researcher, Rice Experimental Pixels Lab Spring 2026

- Engineered a compact, highly directional wind system to synchronize physical airflow with XR environments
- Implemented UDP with networked POE-powered ESP32s, servos, and PWM fans within strict spatial constraints

Technical Projects

wlsn.ee/projects

Batteryless NFC Business Card with MCU & 21-LED Matrix Summer 2026

- Engineered batteryless PCB harvesting 13.56 MHz RF power to drive an MCU and 21 animated LEDs
- Tuned planar loop antenna, inrush-limiting power circuitry, and bare-metal C Charlieplexing system
- Open-source design featured by Hackaday.com, Adafruit Blog, and Microchip; ranked #4 on Hacker News

Custom Rocket Flight Computer & Telemetry Stack 2026

- Designed, programmed, and integrated a custom RP2040-based flight computer in two weeks
- Engineered 1 Hz LoRa telemetry while logging altimeter, accelerometer, and GPS data at 20 Hz
- Built an accompanying ground station with custom antenna to display live data and Google Earth position
- Designed dense 1cm^2 4-layer V2 avionics PCB with STM32L4, dual IMUs, barometer, and flash

UAV with Position-Holding in GPS-Denied Environments 2022 - 2025

- Led national-competition team in design of UAV with optical flow and LiDAR to navigate in tight environments
- Implemented EKF sensor fusion, remote telemetry, modular airframe, and servo-actuated gripper

Doppler & Synthetic Aperture Radar at MIT Lincoln Laboratory RISE Summer 2024

- One of ~4% of applicants selected for DoW-sponsored residential research experience
- Researched, built, and programmed Doppler and synthetic aperture radars, mentored by Lincoln Labs staff

Skills

- **Certifications:** General Class Amateur Radio License, Tripoli Level 1 High Powered Rocketry Certification
- **Hardware & Systems:** Radar/mmWave, LiDAR, UAVs, Microcontrollers, LoRa, SPI/I2C/UART, Networking
- **Software:** C, C++, Python, Altium Designer, KiCad, CAD (Solidworks, Fusion 360, Onshape), ArduPilot, Git, Linux
- **Tools:** Oscilloscopes, Multimeters, SMD Soldering, Function Generators, 3D Printing, Hand Tools